

U THONG AGROMET, Thailand

WMO: 484270

Lat: 14.304N

Lon: 99.865E

Elev: 22

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.6	64.6	52.5	58.8	68.1	56.4	67.9	71.2	10.6	78.5	8.9	76.9	0.5	N/A	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.0	100.5	78.2	98.5	78.3	96.7	78.4	81.9	91.0	81.3	90.5	80.8	90.1	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	155.7	85.1	79.2	151.8	84.4	78.8	149.5	83.9	45.7	91.3	45.0	90.3	44.5	90.4	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.2	7.0	5.3	DB	57.3	102.4	4.4	2.3	54.2	104.0	51.7	105.4	49.2	106.7	46.0	108.4	88.5
			WB	54.9	84.1	4.3	1.8	51.7	85.4	49.2	86.5	46.8	87.5	43.6	88.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	77.4	81.4	85.0	88.0	87.3	85.8	84.7	84.6	84.0	82.9	81.1	77.7
	DBStd	4.39	4.21	3.23	3.72	3.25	2.93	2.08	2.22	1.80	1.82	1.97	2.71	4.10
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12161	849	880	1085	1139	1156	1073	1075	1071	1021	1021	932	859
	CDD65	6686	384	460	620	689	691	623	610	606	571	556	482	394
	CDH74	76134	3438	4977	7682	9224	8890	7609	7115	6917	6293	5913	4712	3364
Wind	WSAvg	0.9	0.8	0.7	0.9	0.9	0.8	1.0	1.2	1.0	0.6	0.4	0.9	1.1
	PrecAvg	42.2	0.3	0.6	1.4	2.6	5.4	4.0	4.3	4.4	9.1	8.2	1.7	0.2
	PrecMax	55.0	2.1	3.6	6.9	9.2	10.9	7.6	7.2	7.2	17.1	14.4	5.2	1.6
	PrecMin	29.3	0.0	0.0	0.0	0.0	0.7	0.5	2.0	2.0	4.1	1.1	0.0	0.0
	PrecStd	6.0	0.4	0.8	1.6	2.1	2.3	1.7	1.6	1.3	3.3	3.2	1.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.9	96.2	101.0	103.7	102.6	98.6	97.0	96.1	95.2	93.9	92.5	92.4
		MCWB	76.7	77.7	76.1	77.0	78.6	79.4	78.5	79.1	79.6	79.8	79.5	77.1
	2%	DB	91.0	94.3	98.3	101.1	99.5	96.1	94.4	93.8	93.4	91.8	91.3	91.0
		MCWB	76.7	77.2	76.5	78.5	78.9	79.3	78.5	78.9	79.6	79.6	78.8	76.9
	5%	DB	89.2	92.7	96.5	99.3	97.3	94.1	92.6	92.1	91.9	90.6	89.9	89.0
		MCWB	75.9	76.6	77.0	78.5	79.3	79.2	78.3	78.7	79.4	79.4	78.0	76.3
	10%	DB	87.1	91.0	94.7	97.0	95.2	92.2	90.9	90.5	90.3	89.3	88.4	86.7
		MCWB	74.8	76.3	77.3	78.5	79.4	79.0	78.1	78.4	79.2	79.2	77.2	75.1
	0.4%	WB	79.8	80.8	81.5	82.3	82.7	82.5	81.0	81.6	82.3	81.9	81.7	79.2
		MCDB	89.3	91.4	93.6	92.9	92.0	92.1	89.6	89.4	90.3	88.3	89.1	88.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	78.4	79.6	80.4	81.4	81.5	81.0	80.2	80.7	81.2	81.0	80.7	78.3
		MCDB	87.9	89.9	91.1	92.0	91.3	90.8	89.1	89.7	89.4	88.1	88.6	87.3
	5%	WB	77.4	78.7	79.6	80.7	80.8	80.3	79.5	79.9	80.5	80.5	79.8	77.4
		MCDB	86.1	88.8	89.8	91.3	90.7	89.9	88.3	88.5	88.8	87.8	87.0	86.0
	10%	WB	76.4	77.8	78.9	80.1	80.2	79.8	78.9	79.2	79.8	79.9	78.8	76.3
		MCDB	84.2	87.3	88.6	90.5	90.0	88.5	87.4	87.2	87.6	86.8	85.3	84.3
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.4	18.8	18.0	15.6	13.8	13.0	13.4	13.3	12.6	13.9	16.5
		MCDBR	20.7	21.1	21.5	20.4	18.3	16.3	15.3	15.2	15.5	14.4	15.9	18.6
		MCWBR	8.0	6.4	5.0	4.4	4.3	4.2	4.1	4.3	4.7	4.7	5.5	6.8
	5% WB	MCDBR	18.9	18.9	18.8	18.3	16.1	14.8	13.8	14.2	14.2	13.5	14.3	17.4
		MCWBR	7.9	6.7	5.4	4.8	4.6	4.3	4.4	4.9	5.1	4.9	5.3	7.1
Clear-Sky Solar Irradiance	taub		0.536	0.605	0.651	0.574	0.490	0.448	0.440	0.441	0.468	0.575	0.521	0.514
	taud		1.953	1.816	1.738	1.963	2.259	2.375	2.397	2.382	2.297	1.949	2.073	2.031
	Ebn at Noon		241	231	224	242	259	267	270	273	267	235	244	244
	Edh at Noon		57	69	75	60	44	39	38	39	43	59	50	52
All-Sky Solar Radiation	RadAvg		1525	1674	1753	1907	1740	1613	1510	1499	1525	1477	1510	1489
	RadStd		91	70	130	108	139	115	105	107	100	93	98	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	+0.38	N/A	N/A	N/A	N/A	+0.47	+0.50	N/A	N/A	+180

Nomenclature: See separate page